

Efficient Out-of-Distribution Generalization for Pretrained GNNs via Prompt Learning

Zhenmeng Zuo , Changsheng Li , Mingkui Tan , Xing Gong , Ye Yuan , and Guoren Wang 

Abstract—Graph out-of-distribution (OOD) generalization enables graph neural networks (GNNs) to perform robustly in applications when confronted with new environments not encountered during training. For better realization of OOD generalization, current methods are increasingly focused on designing more complex causal inference models trained from scratch, without considering the potential of existing pretrained models, and increased computational complexity. In this article, we propose an efficient framework [efficient graph OOD generalization (EGOG)], which uses prompt learning for pretrained GNNs to extract causal subgraphs from input graph data, for enhancing the generalization ability of the pretrained GNNs. Specifically, we design a graph causal prompt module to enhance the generalization ability of the pretrained GNNs. First, we use graph prompt learning with a causal features generator that enables pretrained GNNs to decouple causal features from spurious features in graph nodes. Then, considering that the structure of graph is also affected by distribution shift, we further design a causal structure generator (CSG), which reuses the pretrained GNNs to analyze graph's structure, decoupling causal substructures from spurious substructures. This approach enhances the pretrained GNN's ability to decouple causal subgraph from both the node and structure perspectives. Meantime, we employ gradient reversal layers and information bottleneck to encourage our method to more accurately disentangle causal subgraph from spurious subgraph. Experiment results show that EGOG can be applied to pretrained GNNs to enable generalization ability. Compared with the state-of-the-art baseline, EGOG shows an average relative improvement of 2.73% on two synthetic datasets while using only average 4.38% of the trainable parameters.

Impact Statement—This study introduces EGOG, an efficient framework for OOD generalization in GNNs that leverages graph prompt learning to significantly enhance the generalization ability of pretrained GNNs with minimal training cost. EGOG employs a causal feature generator and a CSG to decouple causal subgraphs

from spurious subgraphs from both node feature and structural perspectives, effectively avoiding environment-related spurious correlations. By integrating information bottleneck and gradient reversal adversarial mechanisms, EGOG achieves competitive performance across multiple synthetic and real-world datasets and substantially reduces the number of trainable parameters. This approach offers a more robust and economical artificial intelligence (AI) solution for real-world applications involving graph data, such as social network analysis, drug discovery, and financial risk control, enabling reliable AI deployment in complex and dynamic environments.

Index Terms—Efficient learning, graph neural networks (GNNs), out-of-distribution (OOD) generalization, prompt learning.

I. INTRODUCTION

GRAPH neural networks (GNNs) have become a widely adopted approach to processing and analyzing unstructured data [1], [2], such as social recommendations [3], drug discovery [4], and fraud detection [5]. However, deployed GNNs often encounter data during inference that significantly differ from the data they were trained on, leading to poor performance in testing on unseen distributions, a challenge known as the out-of-distribution (OOD) problem. Typically, the mismatch between the distributions of training and testing data stems from environmental factors within the collected data (e.g., diverse sharing themes among users across different social platforms). This introduces challenges for the application of GNNs in practice.

To enhance the generalization capability of GNNs and address the challenges in OOD generalization in graphs, many approaches have been proposed from the perspective of data, model design, and training strategies [6]. One significant research direction is invariant learning, which assumes that the information used for prediction can be divided into two parts: invariant subgraph and spurious subgraph. The invariant subgraph maintains a stable relationship with the labels in different environments and can be used to deduce the labels, while the reserved subgraph may change with the environment and mislead the GNNs into learning spurious correlations between the environment and the labels, thus affecting the model's generalization ability. Invariant learning aims to extract stable features that remain unaffected by environmental changes from multiple environments, while ignoring spurious subgraphs, to achieve OOD generalization under distribution shifts.

Invariant learning methods typically require the construction of multiple neural network modules to ensure that the GNNs can

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find invariant subgraphs [7], [8], [9], with all modules needing to be trained from scratch. For example, Gui et al. [7] utilized four encoders during training to fully leverage environmental labels and extract invariant subgraphs, which significantly increases the computational burden and training time.

Recently, pretrained models in computer vision and natural language processing (NLP) have garnered increasing attention due to their efficient performance in transferring to new tasks [10], [11]. By designing appropriate prompts, pretrained models can quickly adapt to new tasks with superior performance, reducing the training resources required for new tasks. Similarly, in the field of GNNs, there are also many works on prompt learning [12], [13] and pretrained works. Additionally, there are numerous pretrained encoders available for reusing. Therefore, inspired by the advancements of prompt learning, which benefits from the efficient task transfer capabilities of pretrained models, we consider enhancing the OOD generalization ability of pretrained GNNs using prompt learning, thereby reducing the training costs for OOD generalization.

However, directly using prompt learning to achieve OOD generalization for pretrained GNNs is unrealistic, and without specific designs for OOD generalization, they can easily establish connections with spurious subgraphs. However, an issue persists:

Some trivial information unrelated to environmental labels may still be retained, affecting the generalization performance of the pretrained GNNs.

In this article, to leverage pretrained GNNs for efficient OOD generalization, we propose our efficient graph OOD generalization framework (EGOG), which uses graph prompt learning to enhance pretrained GNNs OOD generalization ability. First, we design a causal feature generator (CFG) that employs graph prompt learning with pretrained GNNs to identify causal features on graph nodes. This is achieved using an information bottleneck [14], [15] to retain task-label-relevant causal features and decouple spurious environmental features. Second, to address OOD issues in graph structures, we implement a causal structure generator (CSG). This module decouples causal substructures from spurious substructures by reusing the pretrained GNNs, thereby addressing structural OOD issues. By utilizing both the CFG and the CSG, we can decouple an input graph into a causal subgraph and a spurious subgraph. We ensure the quality of this decoupling by employing a gradient reversal layer [16] to encourage invariance during training, ensuring that the causal subgraphs contain no environmental information while the spurious subgraphs contain no task-relevant information. Finally, we evaluate our method on the GOOD dataset [17]. We apply our approach to both synthetic and real-world datasets and demonstrate that it efficiently improves the OOD generalization performance of pretrained GNNs.

Specifically, our contributions are summarized as follows.

- 1) We propose a framework named EGOG, designed to decouple causal subgraphs and spurious subgraphs from graph data, enhancing the generalization ability of pretrained GNNs.

- 2) Through experiments, we demonstrate that EGOG significantly and efficiently improves the OOD generalization performance of pretrained GNNs with the lowest training cost.
- 3) To the best of our knowledge, we are the first to employ prompt learning on pretrained GNNs to address the challenge of OOD generalization.

II. RELATED WORKS

A. OOD Generalization

The goal of OOD generalization is to enhance a model's ability to generalize, enabling it to achieve good results on test sets with unseen data distributions. Methods based on causal inference aim to deduce causal variables relevant to the task from the data for label prediction, operating under the assumption that even if the data distribution changes, the causal relationship between the data and labels remains constant [18]. Methods based on invariant learning operate under the assumption that there are features or representations within the data that remain unchanged across different environments [16], [19], [20], [21]. The OOD problem also arises in unstructured graph data, as highlighted by Li et al. [6]. The unique characteristics of graph data make addressing OOD generalization challenges even more difficult. Explore-to-extrapolation risk minimization [22] maximizes the variance of risks across multiple virtual environments using several explorers, enabling the model to extrapolate from a single observed environment. Discovering invariant rationales (DIR) [8] constructs an interpretable model to select subgraphs as causal patterns and conducts interventions on the training distribution to create multiple interventional distributions. Shift-robust graph neural networks [23] accounts for distributional differences between biased training data and the graph's true inference distribution, adapting GNNs to the presence of distributional shifts between the labeled nodes used for training and the rest of the dataset. Graph invariant learning (GIL) [24] captures invariant relationships between predictive graph structural information and labels in a mixture of latent environments by jointly optimizing three tailored modules. Causality Inspired Invariant Graph LeArning (CIGA) [9] concludes that OOD generalization on graphs is achievable when models focus only on subgraphs containing the most information about the causes of labels. Recently, label and environment causal independence (LECI) [7] has the first to propose a graph-specific strategy that leverages precollected environmental data, developing a domain-adversarial training strategy.

Unlike transfer learning or parameter-efficient fine-tuning methods such as parameter-efficient fine-tuning improves generalization in GNNs [25], which adapt models to new tasks or domains via fine-tuning, OOD generalization focuses on the same task across disjoint distributions. In this work, we adhere to the strict OOD setting, where the model is trained on specific source domains (e.g., specific graph sizes and motifs) and evaluated on completely unseen domains to test intrinsic robustness, distinct from cross-task transferability.

B. Prompt Learning

Prompt learning, an emerging paradigm in the field of NLP and machine learning [26], has garnered significant attention for its potential to leverage the vast knowledge embedded in pretrained language models with minimal task-specific adaptation. Prompts can be crafted through two distinct approaches: manually designed (discrete prompts) or automatically learned via neural networks (soft prompts) [27], [28]. Beyond the realm of NLP, researchers such as Jia et al. [29] and Bahng et al. [30] have extended this approach to the field of computer vision. By prompting each pixel within an image, they aim to improve the efficient adaptability of large-scale pretrained visual models. Zheng et al. [31] explored OOD generalization in visual tasks using prompt learning. They leveraged prompts to learn domain-specific knowledge, and for test data, they utilized a combination of domain-specific prompts as domain cues to achieve model generalization. Recently, in the field of GNNs, prompt learning methods based on graph datasets have gained popularity. Sun et al. [12] introduced prompt learning to GNNs, investigating the transfer learning capabilities of prompt learning in edge prediction tasks. Additionally, Fang et al. [13] proposed the graph prompt feature (GPF), achieving universality in graph prompting by adaptively acquiring prompt graphs for downstream tasks. Sun et al. [32] studied the prompting topic for graphs with the motivation of bridging the gap between pretrained models and various graph tasks. However, the application of prompts for achieving OOD generalization in the graph domain remains underexplored.

III. METHOD

In this section, we introduce our EGOG framework, to enhance pretrained GNNs' OOD generalization through graph prompt learning. The framework is illustrated in Fig. 2. In Section III-A, we present the statement for OOD generalization in graph data, and in Section III-B, we provide an overview of our framework. Then, in Section III-C, we describe the application of graph prompt learning to aid pretrained GNNs in finding causal features in graphs and the use of an information bottleneck as a loss function to ensure that prompts capture causal features within pretrained GNNs. In Section III-D, we present our work on leveraging pretrained GNNs to identify causal substructures, enabling efficient decoupling between causal subgraphs and spurious subgraphs for graph OOD generalization.

A. Problem Formulation

In this section, we investigate the OOD generalization problem for graph data. Let $G = (X, A)$ represent the input graph, where $X \in \mathbb{R}^{n \times d}$ represents the feature matrix of nodes, n denotes the number of nodes, and d represents the feature dimension of each node, $A \in \{0, 1\}^{n \times n}$ is the adjacency matrix of the graph G . If there exists an edge between nodes x_i and x_j , then $A_{ij} = 1$; otherwise, $A_{ij} = 0$. Y represents the target label, while E denotes the environmental label of the graph G from which it was collected.

OOD issues arise when the distribution of the training dataset differs from that of the testing dataset. Specifically, for the

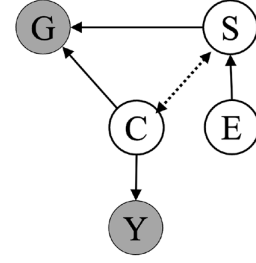


Fig. 1. Structural causal model (SCM) of OOD shift assumptions. Each node denotes a variable, and each directional edge represents causation. Grey nodes denote observable variables.

training dataset $P_{\text{train}}(G, Y)$ and the testing dataset $P_{\text{test}}(G, Y)$, we observe $P_{\text{train}}(G, Y) \neq P_{\text{test}}(G, Y)$. As depicted in Fig. 1, the input data G in OOD generalization tasks comprise a causal part C , which stably exists in different distributions, and a spurious part S . The objective of OOD generalization is to identify the causal part C , enabling the model to trace the $C \rightarrow Y$ pathway. However, the presence of a spurious association path $S \leftarrow C$ may lead the model to learn a spurious pathway $S \rightarrow C \rightarrow Y$, obscuring the true causal relationship and diminishing the model's ability to generalize [8]. Data distribution shifts potentially due to the presence of numerous environment-related spurious features in the data collection environment [7], [33], which exhibits significant variability across different environments E . Therefore, we assume that C denotes causal part that remains consistent across diverse environments, whereas S relates to environmental factors that are spurious and inconsistent. Consequently, we can partition the input data G into two parts: the noncausal, spurious part G_{spu} and the causal part G_{cau}

$$G_{\text{spu}} \rightarrow E, G_{\text{cau}} \rightarrow Y, \quad \text{while } G_{\text{spu}} \perp G_{\text{cau}}. \quad (1)$$

As depicted in (1), conventionally, G is perceived as being composed of G_{cau} and G_{spu} in a complementary manner. The objective of the OOD generalization task is to leverage the environment label E and the task label Y to identify the causal part G_{cau} within the data and to eliminate spurious parts G_{spu} . To improve the OOD generalization ability of GNNs, causal subgraph G_{cau} need to be learned between input data G and target Y from multiple environments, which can adopt the learning strategy of minimizing the empirical risk [8]

$$\min_{h_C, h_Y} = \mathcal{R}(h_C \circ h_Y(G), Y) \quad (2)$$

where $\mathcal{R}(\cdot)$ is the risk function, which can be the cross-entropy loss. h_C, h_Y are two modules, where $h_C : G \rightarrow G_{\text{cau}}$ hopes to discover G_{cau} , $h_Y : G_{\text{cau}} \rightarrow Y$ outputs the prediction to approach Y .

It is noteworthy that as research on OOD generalization progresses, increasingly complex and effective OOD algorithms continue to emerge. Specifically, the complexity of modules h_C and h_Y has increased, inevitably raising the computational costs of training GNNs for OOD generalization. For instance, Gui et al. [7] have proposed the additional utilization of environmental labels E to identify G_{spu} , thereby enhancing the extraction of G_{cau} . However, this approach further complicates

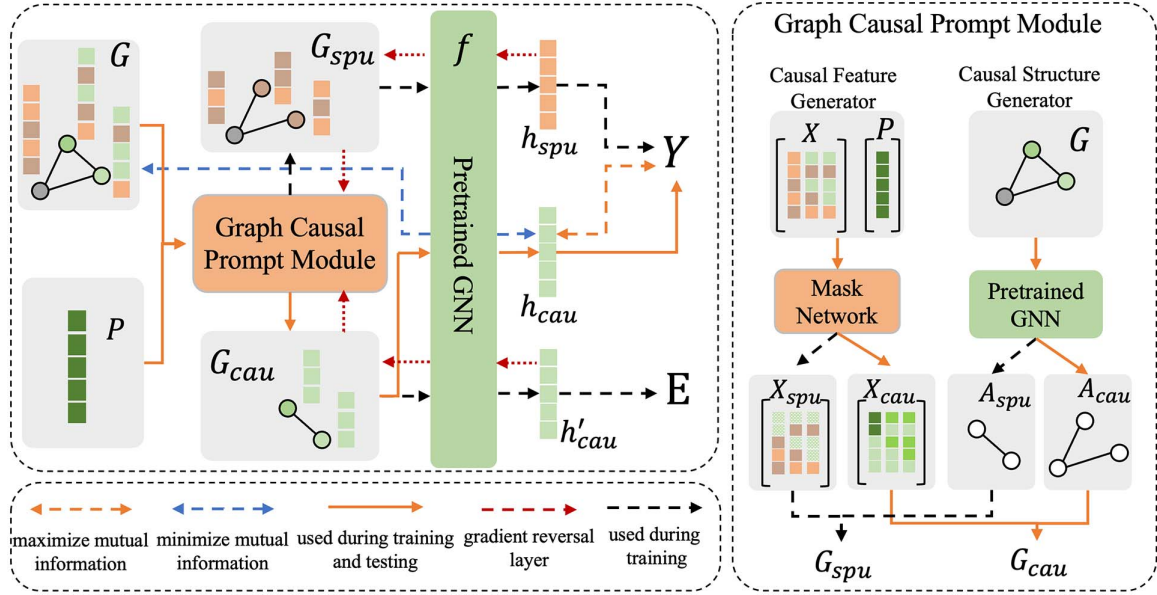


Fig. 2. Framework of the EGOG. We design a GCPM with prompt P to guide pretrained GNN f in discovering and extracting causal features. The GCPM extracts the causal subgraph $G_{\text{cau}} = (A_{\text{cau}}, X_{\text{cau}})$ and the spurious subgraph $G_{\text{spu}} = (A_{\text{spu}}, X_{\text{spu}})$ from the input $G = (X, A)$. To ensure that GCPM can accurately distinguish between G_{cau} and G_{spu} , we employ information bottleneck for causal graph feature $h_{\text{cau}} = f(G_{\text{cau}})$ and gradient reversal layer adversarial for $h_{\text{spu}} = f_{\text{grl}}(G_{\text{cau}})$ and $h'_{\text{spu}} = f_{\text{grl}}(G_{\text{cau}})$. f_{grl} means using gradient reversal layer adversarial for the pretrained GNN f .

the algorithm, potentially leading to increased computational complexity.

B. Overview of Our Framework

To achieve OOD generalization efficiently, we propose a framework that utilizes graph prompt learning, leveraging the abundance of pretrained GNNs available in practice, as illustrated in Fig. 2. The graph causal prompt module (GCPM) employs prompt learning to guide pretrained GNNs in identifying causal features from the nodes and further separates the node features into causal features X_C and spurious features X_S , constrained by an information bottleneck, as detailed in Section III-C. Subsequently, the pretrained GNNs are reused to analyze the structure of the graph G , extracting OOD-related features for each edge and utilizing the Gumbel-Sigmoid [34] to sample causal substructures A_C and spurious substructures A_S . This yields the following:

$$G_{\text{cau}} = (X_{\text{cau}}, A_{\text{cau}}), G_{\text{spu}} = (X_{\text{spu}}, A_{\text{spu}}). \quad (3)$$

To better train the GCPM to decouple the causal subgraph G_{cau} and the spurious subgraph G_{spu} from G , we employ a gradient reversal layer [16] to ensure that each subgraph contains only its respective features, as detailed in Section III-D.

C. GCPM for Causal Features

Node features play a crucial role in capturing the intrinsic properties of each node and their interactions within the graph. By accurately extracting and leveraging these features, we can better understand the underlying patterns and structures that generalize well across different distributions. To achieve this goal, we employ causal prompt learning to identify causal features from graph nodes.

1) *Prompt Learning for OOD Generalization*: By providing the pretrained GNNs with carefully designed prompts, we can guide them to focus on the causal features within the data. This approach leads to improved robustness and performance when the model encounters new, unseen data distributions. Referring to existing works on graph prompt learning, we utilize a pretrained GNN model f with fixed parameters and a trainable decoder with parameters θ . The prompt with parameters P operates on the input graph data, optimizing P and θ to maximize the likelihood of predicting the correct labels Y without tuning the pretrained model f . This can be formulated as follows:

$$\max_{P, \theta} \mathbb{P}_{P, \theta}(y | f(G, P)). \quad (4)$$

In this work, we reference the graph prompt learning method GPF-Plus [13], which exhibits stronger universality. Specifically, for the node features $X = \{x_1, x_2, \dots, x_n\}$ in the graph data G , we can provide prompt for the features of each node in the graph by designing a set of learnable vectors $\hat{P} = \{p_1, p_2, \dots, p_\Gamma\}$ and Γ learnable linear projections a

$$X^* = X + P = \{x_1 + p_1^*, x_2 + p_2^*, \dots, x_n + p_n^*\} \quad (5)$$

$$P = \left\{ p_i^* = \sum_{j=1}^{\Gamma} \alpha_{i,j} p_j \right\}_{i=1}^n, \quad \alpha_{i,j} = \frac{\exp(a_j^T x_i)}{\sum_{l=1}^{\Gamma} \exp(a_l^T x_i)} \quad (6)$$

where Γ is a hyperparameter representing the number of learnable vectors $\hat{P}$, X^* represents the features after prompting, and P is the prompt vector.

However, simply using prompts can still lead the pretrained GNNs to fall into the trap of spurious correlations, which requires careful design to ensure that the prompts guide the model in discovering causal features.

2) *CFG*: Directly applying prompts to the input graph still retains spurious features on each node. On the one hand, it affects the subsequent extraction of causal features by the pretrained GNN f , and on the other hand, it necessitates the decoupling of spurious node features during the exploration of graph causal structures. Therefore, we designed a CFG to further use prompts to decouple causal feature from the node features. To achieve this, we take the node features X and the prompt P as inputs, fuse them through one MLP layer, and then apply a sigmoid function to obtain a soft mask to enhance the causal feature within X

$$\text{mask} = \text{Sigmoid}(\text{mlp}_{\text{mask}}(P + X)) \quad (7)$$

$$X_{\text{cau}} = \text{mask} \cdot X + P, \quad X_{\text{spu}} = (1 - \text{mask}) \cdot X. \quad (8)$$

3) *Information Bottleneck for Causal Prompt Learning*: To better enable prompts to identify causal features in nodes and assist pretrained GNN in uncovering these causal features, effective supervision of the prompts is necessary. To solve this problem, we employ information bottlenecks to discover causal features. The goal of the information bottleneck is to identify the appropriate intermediate feature h , which aims to minimize the following formula:

$$\min I(h; G) - \beta I(h; Y). \quad (9)$$

Here, $I(h; G)$ represents the mutual information between h and G , and $I(h; Y)$ denotes the mutual information between h and Y . Considering that mutual information is nonnegative, (9) is designed to compress h to have less information about G while retaining essential information about Y .

It is computationally intractable to directly optimize (9). Therefore, we use CLUB [14] and MINE [15] to estimate the upper and lower bounds of mutual information, respectively. To achieve this, we can use the causal subgraph $G_{\text{cau}} = (X_{\text{cau}}, A_{\text{cau}})$ obtained from GCPM. The features h_{cau} are derived from a pretrained GNN f and serve as the intermediate feature in the information bottleneck. Here, X_{cau} is generated by the CFG, and the method to obtain A_{cau} will be discussed in Section III-D. Therefore, in combination with the optimization objectives of information bottlenecks, we have the following:

$$\mathcal{L}_{ib} = I_{\text{club}}(h_{\text{cau}}; G) - \beta I_{\text{mine}}(h_{\text{cau}}; Y). \quad (10)$$

In our experiment, the hyperparameter β is set to 0.5.

D. GCPM for Causal Subgraph

1) *CSG*: To obtain causal substructures, we designed a CSG. It can analyze the structure of the input graph G and samples two substructures: causal substructures and spurious substructures by reusing pretrained GNN. First, the pretrained GNN, with prior knowledge, can extract structural information from the input graph without requiring training. We use this pretrained GNN with an additional mlp_{str} to extract the structure features of the graph data. After that, following the work of [7], we use Gumbel-Sigmoid [34] to sample the causal substructure and the spurious substructure. The formula is as follows:

$$\begin{aligned} A_{\text{cau}} &= \text{Gumbel}(\text{mlp}_{\text{str}}(f(G))) \\ A_{\text{spu}} &= A - A_{\text{cau}}. \end{aligned} \quad (11)$$

2) *Domain-Adversarial Training for GCPM*: Using the CFG and the CSG modules, we can separate causal features X_{cau} and spurious features X_{spu} , as well as causal substructures A_{cau} and spurious substructures A_{spu} . By combining them, we can obtain the causal subgraph $G_{\text{cau}} = (X_{\text{cau}}, A_{\text{cau}})$ and the spurious subgraph $G_{\text{spu}} = (X_{\text{spu}}, A_{\text{spu}})$.

To better train the GCPM to decouple the causal subgraph G_{cau} and the spurious subgraph G_{spu} from G , we employ a gradient reversal layer [16] as domain-adversarial method to ensure that each subgraph contains only its respective features. Specifically, for the causal subgraph G_{cau} , we use an MLP layer, $\text{mlp}_{\text{adv}E}$, to predict its environment label E . For the spurious subgraph G_{spu} , we use another MLP layer, $\text{mlp}_{\text{adv}Y}$, to predict the classification label Y

$$\mathcal{L}_{\text{adv}Y}(\lambda_1) = \mathbb{E}[\log P(Y|\text{mlp}_{\text{adv}Y}(f(G_{\text{spu}})))]$$

$$\mathcal{L}_{\text{adv}E}(\lambda_2) = \mathbb{E}[\log P(E|\text{mlp}_{\text{adv}E}(f(G_{\text{cau}})))] \quad (12)$$

$$\begin{aligned} \text{Grad}_{\text{mlp}_{\text{adv}Y}} &= \frac{\partial \mathcal{L}_{\text{adv}Y}}{\partial \theta_{\text{mlp}_{\text{adv}Y}}}, \quad \text{Grad}_{G_{\text{spu}}} = -\lambda_1 \frac{\partial \mathcal{L}_{\text{adv}Y}}{\partial G_{\text{spu}}} \\ \text{Grad}_{\text{mlp}_{\text{adv}E}} &= \frac{\partial \mathcal{L}_{\text{adv}E}}{\partial \theta_{\text{mlp}_{\text{adv}E}}}, \quad \text{Grad}_{G_{\text{cau}}} = -\lambda_2 \frac{\partial \mathcal{L}_{\text{adv}E}}{\partial G_{\text{cau}}}. \end{aligned} \quad (13)$$

As shown in (12) and (13), $\text{mlp}_{\text{adv}E}$ is supervised by the environment label E to identify environment-related information from G_{cau} , while $\text{mlp}_{\text{adv}Y}$ is supervised by the graph label Y to identify task-related information from G_{spu} . Simultaneously, the gradient reversal layer performs domain-adversarial training by reversing the gradient during backpropagation, thereby encouraging G_{cau} to refine causal features that fool $\text{mlp}_{\text{adv}E}$, and encouraging G_{spu} to refine spurious features that fool $\text{mlp}_{\text{adv}Y}$.

3) *Training Strategy*: In Sections III-B–III-D, we have detailed how our proposed framework efficiently utilizes pretrained GNNs to extract the causal subgraph G_{cau} and provided the loss optimization objectives. The training objectives are shown as follows:

$$\mathcal{L}_{\text{all}} = \mathcal{L}_{\text{task}} + \alpha \mathcal{L}_{ib} + \mathcal{L}_{\text{adv}Y}(\lambda_1) + \mathcal{L}_{\text{adv}E}(\lambda_2). \quad (14)$$

Here, $\mathcal{L}_{\text{task}} = \ell(Y, G, P)$ represents the task loss for predict label Y , $\mathcal{L}_{ib}$ (10) uses the information bottleneck to constrain prompts to the pretrained GNN to extract causal features from nodes with hyperparameter α , and $\mathcal{L}_{\text{adv}E}$ and $\mathcal{L}_{\text{adv}Y}$ (13) employ domain-adversarial learning to decouple the causal subgraph G_{cau} from the spurious subgraph G_{spu} within the input G . λ_1 and λ_2 are two hyperparameters for gradient reversal layers (13).

The trainable parameters in this work are much fewer compared to other works. In our framework, the trainable parameters include the prompt P , the two MLPs mlp_{mask} and mlp_{str} in GCPM, the decoder for task classification, and the domain-adversarial training decoders $\text{mlp}_{\text{adv}E}$ and $\text{mlp}_{\text{adv}Y}$.

IV. EXPERIMENTS

A. Experimental Setup

1) *Dataset*: To evaluate our method, we selected four datasets from the GOOD benchmark [17]. These include two synthetic datasets, GOODCMNIST and GOODMotif, and two

TABLE I
PERFORMANCE OF OOD GENERALIZATION ON GOODMOTIF AND GOODCMNIST DATASETS

Method	GOODMotif-Basis		GOODMotif-Size		GOODCMNIST-Color	
	Trainable Params ↓ (K)	Accuracy ↑	Trainable Params ↓ (K)	Accuracy ↑	Trainable Params ↓ (K)	Accuracy ↑
ERM	910(39%)	60.93(11.11)	910(39%)	56.63(7.12)	2001(96%)	26.64(2.37)
GPF	15	62.21(1.29)	15	55.82(0.32)	12	28.43(0.72)
IRM	910(39%)	64.94(4.85)	910(39%)	54.52(3.27)	2001(96%)	29.63(2.06)
Coral	910(39%)	61.95(10.36)	910(39%)	55.80(4.05)	2001(96%)	29.21(6.87)
DANN	911(39%)	50.62(4.71)	911(39%)	46.61(3.78)	2003(96%)	27.86(5.02)
GroupDRO	910(39%)	68.24(1.94)	910(39%)	51.95(2.80)	2001(96%)	29.07(2.62)
DropNode	910(39%)	75.25(4.24)	910(39%)	53.92(2.79)	2001(96%)	32.86(1.02)
DropEdge	910(39%)	43.92(1.13)	910(39%)	46.22(1.32)	2001(96%)	26.32(0.74)
Mixup	910(39%)	70.29(2.13)	910(39%)	52.39(2.43)	2001(96%)	27.36(1.83)
DIR	911(39%)	34.39(2.02)	911(39%)	43.11(2.78)	2368(96%)	22.53(2.56)
GSAT	1991(72%)	62.27(8.79)	1991(72%)	50.03(5.71)	3083(97%)	35.02(2.78)
CIGA	911(39%)	37.81(2.42)	911(39%)	51.87(5.15)	2368(96%)	25.06(3.07)
OOD-GNN	911(39%)	61.10(7.87)	911(39%)	52.61(4.67)	2368(96%)	26.49(2.94)
OOD-GCL	-	76.23(4.89)	-	58.43(3.15)	-	41.29(3.85)
LECI	6358(91%)	84.56(2.22)	8173(93%)	71.43(2.96)	12704(99%)	51.80(2.70)
EGOG	<u>552</u>	87.49(2.41)	<u>552</u>	<u>65.56(3.5)</u>	<u>90</u>	62.94(2.37)

Note: Here, we present the trainable parameters of OOD algorithms (with the reduction rate of our EGOG shown in parentheses) along with the OOD results from 3 runs on the test datasets (standard deviations are shown in parentheses). The best and second-best results are indicated in bold and underlined, respectively.

real-world datasets, GOODSST2 and GOODTwitter. GOODCMNIST is a semiartificial dataset, where the distribution shift primarily occurs in the node features. It is created by transforming the handwritten digits dataset MNIST [35] into graph data using superpixel techniques [36]. In the GOODCMNIST dataset, the color of the digits serves as the domain. GOODMotif is a synthetic dataset inspired by Spurious-Motif [8]. Each sample in this dataset is composed of a base graph (which can be a wheel, tree, ladder, star, or path) and a motif graph (which can be a house, cycle, or crane), with the motif graph determining the graph label. The type and size of the base graph are chosen as the domains, referred to as GOODMotif-base and GOODMotif-size, respectively. GOODSST2 and GOODTwitter are two sentiment analysis datasets based on real-world natural language. Each sentence is converted into a grammar tree graph where individual words serve as nodes, with their associated word embeddings acting as node features. Sentence lengths are selected as the domains for GOODSST2 and GOODTwitter.

2) *Baseline Methods*: We first employ empirical risk minimization (ERM) as a foundational baseline to evaluate the performance without OOD generalization constraints. Subsequently, we validate our method against a comprehensive set of baselines covering general, augmented, and graph-specific approaches. We first compare our methods with ERM and GPF [13]. For general OOD generalization, we selected IRM [20], VREx [21], DANN [16], GroupDRO [37], and Coral [19]. To assess general OOD strategies applied to graphs, we include graph data augmentation methods such as DropNode [38], DropEdge [39], and Mixup [40]. These are compared alongside state-of-the-art graph-specific OOD algorithms: DIR [8], GSAT [41], CIGA [9], OOD-GNN [42], OOD-GCL [43],¹ and LECI [7].

¹Since OOD-GCL is not open source, we only report the dataset accuracy mentioned in the article.

3) *Implementation Details*: In our experiments, we utilize a pretrained GNN and follow the settings in GraphCL [44], employing GIN [45] as the GNN backbone, and adopting node dropping as the data augmentation strategy for contrastive learning training. To maximize the information captured by the prompts, we apply GPF-Plus [13] with hyperparameter $\Gamma = 10$ for prompt learning after the first layer of the GNN. On the GOODMotif and GOODCMNIST dataset, hyperparameters we set α to 0.1 and 0.2, respectively, set β to 0.5 for both datasets. We set λ_1 and λ_2 to 0.1 for both datasets. Our implementation is under the architecture of PyTorch [46] and PyG [47], with graphics cards NVIDIA RTX 2080ti.

B. Performance Comparison

We first utilize the synthetic datasets GOODMotif and GOODCMNIST to explore the efficiency of our method in OOD generalization tasks, with results shown in Table I. On the GOODMotif dataset, our method significantly reduces the number of trainable parameters by 91% compared to the current state-of-the-art method LECI [7], while still maintaining superior performance. This reduction is attributed to our method's use of pretrained GNNs, which can effectively identify causal substructures. For the GOODMotif-size dataset, LECI, which uses more parameters, better identifies spurious correlations between environments and distributions, thereby achieving better performance. Although our method lags behind LECI, it still surpasses other methods and achieves the second-best result while maintaining training efficiency. On the GOODCMNIST dataset, our method outperforms LECI by 11.14%, establishing a new state-of-the-art for this dataset. We believe this is primarily because the GOODCMNIST dataset focuses on distribution shifts in node features, and the utilization of prompt learning directly on each node in the graph makes it easier for prompts to

TABLE II
PERFORMANCE ON GOODSST2 AND GOODTWITTER DATASETS

Method	GOODSST2			GOODTwitter		
	Trainable Params ↓ (K)	ID Accuracy ↑	OOD Accuracy ↑	Trainable Params ↓ (K)	ID Accuracy ↑	OOD Accuracy ↑
ERM	1732(79%)	78.37(2.64)	80.41(0.69)	1733(78%)	54.93(0.96)	57.04(1.70)
GPF	15	88.43(1.32)	81.02(0.48)	15	58.83(0.32)	56.21(1.02)
IRM	1732(79%)	79.73(1.45)	80.17(1.52)	1733(78%)	55.27(1.19)	57.72(1.03)
VREx	1732(79%)	79.31(1.40)	80.33(1.09)	1733(78%)	56.46(0.93)	56.37(0.76)
Coral	1732(79%)	78.24(3.26)	80.97(1.07)	1733(78%)	56.57(0.42)	56.14(1.76)
DANN	1734(79%)	78.74(0.82)	80.36(0.61)	1736(78%)	55.52(1.27)	55.71(1.23)
GroupDRO	1734(79%)	89.59(0.09)	79.21(1.02)	1736(78%)	59.21(0.32)	56.84(0.63)
DropNode	1732(79%)	89.91(0.42)	81.21(1.60)	1736(78%)	58.90(1.23)	56.82(1.30)
DropEdge	1732(79%)	90.30(0.18)	78.93(1.34)	<u>1736(78%)</u>	59.19(1.50)	56.93(1.08)
Mixup	1732(79%)	89.78(0.20)	80.88(0.60)	<u>1736(78%)</u>	58.97(1.54)	56.07(1.64)
DIR	2096(82%)	77.65(0.71)	81.50(0.55)	2097(82%)	55.32(1.85)	56.81(0.91)
GSAT	2814(87%)	78.51(2.26)	80.44(0.59)	2814(87%)	56.10(2.65)	56.37(1.30)
CIGA	2096(82%)	80.37(1.46)	81.20(0.75)	2097(82%)	57.51(1.36)	<u>57.19(1.15)</u>
OOD-GNN	1734(79%)	78.32(3.28)	80.70(1.24)	1736(78%)	56.22(2.02)	56.45(2.16)
LECI	9303(96%)	82.93(0.22)	83.44(0.27)	9303(96%)	59.35(1.44)	59.64(0.15)
EGOG	<u>371</u>	<u>90.02(2.20)</u>	80.97(1.34)	<u>374</u>	59.57(1.19)	57.06(1.10)

Note: Here, we present the trainable parameters of OOD algorithms (with the reduction rate of our EGOG shown in parentheses) along with the OOD results from 3 runs on the test datasets (standard deviations are shown in parentheses). The best and second-best results are indicated in bold and underlined, respectively.

guide the pretrained GNN models in extracting causal features at the node level.

We conduct experiments on the real-world datasets GOODTwitter and GOODSST2, with results shown in Table II. First, our method achieves the best performance on ID testing and reduces the parameter count by over 78%, while also demonstrating comparable performance on OOD.

C. Ablation Study

Subsequently, we conduct ablation experiments on the GOODMotif-base to verify the performance of the modules we propose, where “CSG” is the causal structure generator and “CFG” is the causal feature generator. The results are shown in Table III. First, we assess the performance of the pretrained GNN by fine-tuning the decoder using a pretrained GNN with frozen parameters, without using our proposed method. The results indicate that simply using the pretrained GNN does not achieve OOD generalization performance. We hypothesize that this is because the decoder can still learn some environment-specific spurious information. Next, we add the CSG module to capture the causal structure in graph data. This results in a 44.53% improvement compared to only fine-tuning the decoder, indicating that using guidance on structure significantly enhances the pretrained GNN’s ability to capture causal substructures. Additionally, using the CFG provides a further 2.87% improvement. Considering that our prompt is applied after the first layer of the pretrained GNN, at which point nodes already have some structural features, the CFG can still be effective on GOODMotif, which primarily involves distribution shifts in structure.

D. Hyperparameter Sensitivity

We evaluated the algorithm’s sensitivity to the information bottleneck hyperparameters (α , β) and the gradient reversal

TABLE III
ABLATION STUDY ON
GOODMOTIF-BASE

CSG	CFG	OOD	ID
		39.27	91.60
✓		83.80	92.33
✓	✓	86.67	91.87

layer hyperparameters (λ_1 , λ_2) on the GOODMotif-base dataset (Fig. 3). We observed that $\alpha = 0.1$ yields the best accuracy, while the performance is relatively robust to changes in β between 0.1 and 1. For λ_1 , moderate values around 10^{-1} yield the highest accuracy, while very high values significantly degrade performance. Similarly, for λ_2 , small to moderate values maintain high accuracy, but larger values lead to a noticeable decline. These findings highlight the critical importance of carefully tuning λ_1 and λ_2 to achieve optimal model performance.

E. Training Without Environment Labels

Furthermore, considering that the environment label E is difficult to obtain, we use K-means to generate pseudolabels for the environment. Specifically, we use a pretrained GNN to extract graph features and use K-means clustering to form K classes to obtain pseudolabels for the environment. The model is then trained, and the results are shown in Table IV. Using K-means strategy achieves competitive results compared to environment labels, with performance gaps of only 3.0% on GOODMotif-basis, 4.30% on GOODMotif-size, and 1.56% on GOODCMNIST. This demonstrates the viability of unsupervised environment construction.

F. Computational Efficiency Analysis

Table V quantifies the computational efficiency of different adaptation strategies under identical experimental conditions

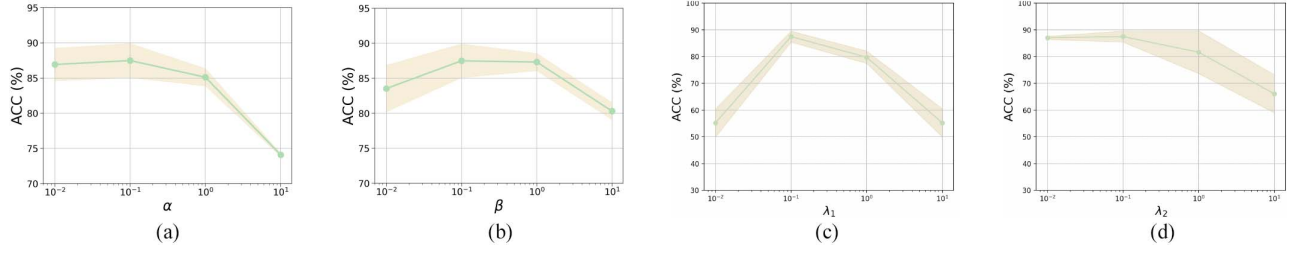


Fig. 3. Parameter sensitivity on α , β , λ_1 , and λ_2 . We conduct the experiments three times under the same conditions. The green line represents the mean values, while the shaded areas indicate the standard deviation. (a) Sensitivity analysis of information bottleneck hyperparameter α . (b) Sensitivity analysis of information bottleneck hyperparameter β . (c) Sensitivity analysis of gradient reversal hyperparameter λ_1 . (d) Sensitivity analysis of gradient reversal hyperparameter λ_2 .

TABLE IV
OOD GENERALIZATION PERFORMANCE (%)
WITHOUT ENVIRONMENT LABELS ON GOODMOTIF
AND GOODCMNIST

Method	Motif-Basis	Motif-Size	CMNIST
Env-label	87.49	65.56	62.94
K-Mean(K = 3)	82.29	60.97	60.23
K-Mean(K = 5)	84.40	61.26	60.92
K-Mean(K = 7)	83.21	60.32	61.38

TABLE V
COMPARISON OF TRAINING AND
INFERENCE LATENCY PER BATCH
(BS = 64) ON GOODCMNIST

Method	Latency (ms)	
	Training	Inference
LECI	195.13	96.09
Full fine-tuning	60.84	23.49
EGOG	54.37	22.74

(batch size = 64, NVIDIA RTX2080ti GPU). EGOG achieves the lowest latency in both training (54.37 ms) and inference (22.74 ms) phases, demonstrating substantial efficiency gains over full fine-tuning and LECI. This confirms that EGOG is an efficient framework with prompt tuning.

V. CONCLUSION

In this work, we introduce an EGOG framework that enhances the generalization ability of pretrained GNNs. By leveraging prompt learning, our approach guides pretrained GNNs to extract causal subgraphs from graph data, addressing the computational complexity challenges of current methods that rely on complex causal inference models. The GCPM, comprising the CFG and the CSG, effectively decouples causal subgraphs from spurious subgraphs from both node features and graph structure perspectives. The incorporation of information bottleneck techniques and a gradient reversal layer ensures the successful separation of causal subgraphs from spurious subgraphs. Experimental results confirm that our proposed framework can improve the generalization of pretrained GNNs while reducing training costs.

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